

Avoiding Loosely Connected Words and Phrases

Some student writers hand in reports made up of paragraphs consisting of one to three sentences only, and others do not paragraph at all. Both of these groups of writers have failed to distinguish between the topics requiring elaboration and the supporting details. The best way out of the dilemma is to make a preliminary outline of the principal points and list below each point the details needed for amplification.

1. Monotonous Repetition:

Nowhere does a poverty-stricken vocabulary betray itself more unpleasantly than in the close proximity of two or more similar words. Surely, with ingenuity, the writer can find a way around such expressions as the following: *"Then. . . Then. . . Then. . . Then,"* when describing a process. A preferable approach could be, *"First. . . Secondly. . . etc.,"* or, *"First. . . Next. . . Then. . . Finally."*

Poor

*A voltage of 7500 volts
gas dried in drying towers
blown by blowers
fans that fan the gases
pumped by pumps
cooled in the cooler*

Better

*a potential of 7500 volts
moisture removed in drying towers
forced along by blowers
fans that propel the gases
forced out by pumps
the heat was dispelled by the cooler*

2. Lack of precision in word choice:

Because engineering reports must convey precise and accurate information, the student should avoid the so-called "wastebasket" words, which are so comprehensive and yet so indefinite. In this category come such verbs as get, use, find, do, put, and, especially, to be.

"This work was done with the aid of several computers, but was done in the main by. . . ." Why not accomplished or executed?

"The bob was put into the water." Why not dropped, lowered, immersed, let down, released into, allowed to fall, made to descend?

"Solving the equation gives. . . ." Why not results in, establishes a value of, yields the expression, reduces to?

3. Misspelled Words:

Bad spelling is one of the most inexcusable faults of which the writer can be guilty. Improperly spelled words are the first things that strike the eye. The reader immediately assumes that the botched spelling is a criterion of the writer's technical skill and that his/her work as an engineer will be no more reliable.

The following words are among those most commonly misspelled in chemical engineering reports, with "separate" the most frequent offender. Most word-processing programs have spell checkers, and failure to use them can result in a poor grade

<i>absorb, absorption</i>	<i>dependence</i>	<i>occur, occurred</i>
<i>accommodate</i>	<i>desiccant</i>	<i>omit, omitted</i>
<i>achieve</i>	<i>elute, eluent or eluant</i>	<i>omission</i>
<i>ambience or ambiance,</i>	<i>empirical</i>	<i>personnel</i>
<i>ambient equip,</i>	<i>equipped</i>	<i>Poiseuille</i>
<i>analyze</i>	<i>equilibrium</i>	<i>proceed, procedure</i>
<i>aqueous</i>	<i>extension</i>	<i>propellant (preferred)</i>
<i>Avogadro</i>	<i>forty or</i>	<i>propellent</i>
<i>begin, beginning</i>	<i>fluctuate</i>	<i>pseudo</i>
<i>believe</i>	<i>fluorine</i>	<i>rarefy</i>
<i>Bernoulli</i>	<i>fluorescent</i>	<i>refer, referred</i>
<i>buoyant</i>	<i>gas, gases</i>	<i>recommend</i>
<i>canceled or cancelled</i>	<i>gauge or gage</i>	<i>Reynolds number (no')</i>
<i>changeable</i>	<i>labeled or labelled</i>	<i>rotameter</i>

a. In present-day practice, certain words are still spelled exclusively with "-er" and others with "-or." Some may be spelled both ways:

"-er" only	"-or" only	Both
<i>condenser</i>	<i>agitator extractor</i>	<i>computer, -or</i>
<i>controller</i>	<i>calculator integrator</i>	<i>connector, -or</i>
<i>digester</i>	<i>compressor moderator</i>	<i>converter, -or</i>
<i>experimenter</i>	<i>detector resistor</i>	<i>conveyer, -or</i>
	<i>evaporator</i>	<i>distributor, -or</i>

b. Avoid simplified spelling in any form of serious writing. Do not use *enuf*, *lite*, *nite*, *tho*, *thot*, *thru*, etc.

PUNCTUATION ERRORS

First and foremost is the rule that, when two or more words modify another word as a unit, the hyphen must be used to show the compound relationship.

In the examples below, chosen because they are among the most commonly used expressions in chemical engineering reports, observe that, without the hyphen, the precise meanings of the modified nouns would be seriously in doubt.

The ordinary reader might not be able to detect the real meanings at all, and even the technician might be puzzled. The writer owes the reader the courtesy of making his/her meaning unmistakably clear, a properly placed hyphen being one way of doing so.

acetic-acid water system
bubble-cap tray
bulk-air temperature
chemical-process equipment
constant-head tank
constant-pressure theory
continuous-stirred-tank reactor
fluid-friction factor
laminar-velocity profile
low-energy bond

gas-bubble column
heat-exchange-product stream
heat-transfer-film coefficient
impact-velocity feed line
liquid-temperature air-enthalpy plot
logarithmic-mean temperature
mass-transfer packed-tower theory
time-drying-rate plots
tray-type column
tube-flow viscometer

a. In a series of hyphenated words having a common base, use suspended hyphens:

a two- or three-year study
dextro- and levo-tartaric acid
low-, medium-, and high-density polyethylenes
T- and H-shaped rib design
twenty-six or -seven days

b. If the preceding adjective is in the comparative or the superlative degree, no hyphen is used:

a high-priced unit	a higher priced unit
a small-size pipe	the smallest size pipe

c. Since a different meaning may be conveyed by writing a prefix solid with the word, a hyphen may have to be used:

resort to other means	re-sort the particles
recover the effluent	re-cover the receptacle
recollect past errors	re-collect the filtrate
microin (correct)	micro-in (better)
reform the methods	re-form the molecules
unionized labor	un-ionized atoms

CAPITALIZATION

- Capitalize all proper nouns such as, names of persons, places, companies, institutions, etc. and the adjectives derived therefrom.
- Capitalize the first word of every sentence and also the first word of every direct quotation.
- When writing such terms as "Reynolds number," "Fanning friction factor," "Newtonian fluid," etc., capitalize the name of the person associated with the concept but not the expression following.
- Capitalize north, east, south, and west only when these words refer to places, not when they refer to directions. Western Phosphates, Inc., sells its products mostly in the Intermountain West. (Place) We traveled west to visit this plant. (Direction).

- e. Do not capitalize the names of elements, chemicals, or their compounds. (Chemical symbols are obvious exceptions.) Names of elements followed by their atomic numbers should be hyphenated. If the elements are referred to by symbol only, the initial letter should be capitalized.

magnesium-12	Mg-12
uranium-238	U-238
chlorine-17	Cl-17

- f. Numbered items should be capitalized when referred to in the text of the report. Write the numbers as figures, not as words.

Equation (7)	Group 2
Chapter 1	Project 3A
Reference 3	Particles 6-9
Table III	Run No. 4

GRAMMATICAL AND MISCELLANEOUS ERRORS

- a. Subject and verb must agree in number; a singular subject requires a singular verb; and a plural subject, a plural verb. (In this connection, remember that the letter "s" indicates the plural of most nouns but the third-person singular of verbs, as: one equation (s.), two equations (pl.); but, the law states (s.), the laws state (pl.).)
- b. A compound subject is plural and takes a plural verb. For example, John and William, Ram and Shayam, The boy and girl, I and You, He and She, etc.
- c. Do not be confused by a "delayed" subject, or a subject which comes after the verb. Off the tops of the distillation columns *come* the light hydrocarbons. (Even though "hydrocarbons" follows the verb, it is nevertheless the subject of the sentence; and the verb should therefore be plural.)
- d. In scientific writing, certain Latin and Greek words still retain their original plural, regardless of informal or conversational use. Be especially on guard with respect to the Latin plural data. Use these and those, not this, with the plurals listed.

Singular	Plural	Singular	Plural
axis	axes	nucleus	nuclei
basis	bases	phenomenon	phenomena
criterion	criteria	quantum	quanta
datum	data	radius	radii
hypothesis	hypotheses	stratum	strata
medium	media		

These data assure the validity of the findings. (Pl.)

Many data were taken. (Pl.)

The datum point was set at three degrees. (S.)

The packing medium was ranching rings. (S.)

The porous media were glass beads and aluminum particles. (Pl.)

- e. When referring to an industrial company or organization, use the singular verb and pronoun.

The company has sent out its catalog.

Johns-Manville manufactures roofing.

General Motors, which invites us. . . .

- f. In technical writing, use only those words accepted on the level of present-day written English. Avoid those marked in the dictionary as colloquial or slang. "Speech may be relatively vague and loose, but writing should be firm and exact."--A. O. Guthrie. Do not use;

"good" for "valid," "accurate," "excellent"

"lab" for "laboratory"

"fix" for "repair"

"in good shape" for "in good condition"

"the reason is because" for "the reason is that"

"most all of" for "almost all"

"rig" for "apparatus"

"plus" for "together with"

"around 300 degrees" for "about" or "approximately"

"to try and distill" for "to try to distill"

"checked into" for "investigated"

- g. Use *should* rather than *would* to express expectation.

This restriction on the use of the equipment *should* make it possible to build up the basic analog in one panel.